Fair HYB-MS Natural-Sampling GAN Experiment

Same 363,692 real URLs • natural sampling • seeds 42/43/44

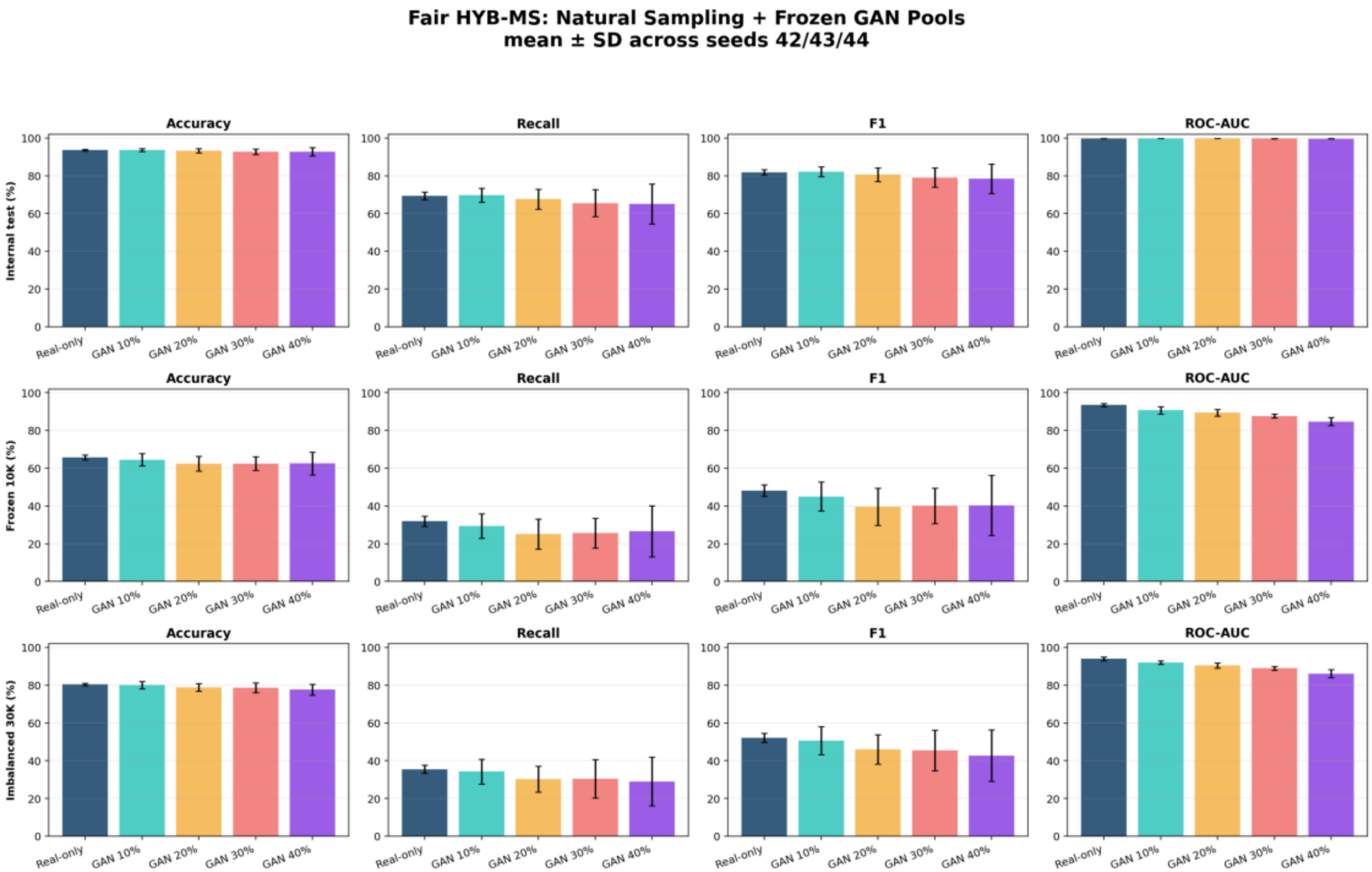
Outcome: real-only transfers best. GAN 10% provides only a tiny internal F1 increase, while all GAN conditions reduce external ranking and threshold performance. Higher synthetic shares worsen the decline, indicating synthetic-to-real distribution mismatch rather than useful new coverage.

Internal: best F1 = GAN 10% (0.8208); ROC-AUC 0.9971

External 10K: best F1 = Real-only (0.4820); ROC-AUC 0.9351

External 30K: best F1 = Real-only (0.5213); ROC-AUC 0.9399

External labels were evaluation-only; no threshold was selected from the 10K or 30K test sets.


Fair HYB-MS: Natural Sampling + Frozen GAN Pools
mean ± SD across seeds 42/43/44
Internal test (%)
Frozen 10K (%)
Imbalanced 30K (%)
Accuracy
Recall
F1
ROC-AUC
Real-only
GAN 10%
GAN 20%
GAN 30%
GAN 40%
Same 363,692 real URLs in every condition • no source-weighted resampling • registered robust thresholds